

Solving Linear Systems by Elimination

Scaling one equation, scaling both equations, and modeling with systems

Directions

- Solve every system by **elimination**. Multiply one or both equations by a constant first so that one pair of like terms becomes opposites (or matches).
 - Show the scaled equations you actually added or subtracted.
 - Write each answer as an ordered pair, or as a labelled pair of values for the word problems.
 - Check your solution in *both* original equations before writing it down.
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1. Solve by elimination.

$$3x + 2y = 16 \quad \text{and} \quad 5x - 4y = 12$$

Answer: _____

2. Solve by elimination.

$$4x + 3y = 18 \quad \text{and} \quad 2x - y = 4$$

Answer: _____

3. Solve by elimination. Neither pair of like terms matches yet, so scale *both* equations.

$$3x + 4y = 25 \quad \text{and} \quad 2x + 3y = 18$$

Answer: _____

4. The drama club sells two kinds of tickets. Three adult tickets and two student tickets cost \$37. Five adult tickets and four student tickets cost \$65. Let a be the price of one adult ticket in dollars and let s be the price of one student ticket in dollars.

(a) Write a system for the two sales and solve it by elimination. Give both prices in dollars.

Answer: _____

- (b) Explain why multiplying the first equation by 2 is enough to eliminate s , with no second multiplication needed.

5. Solve by elimination.

$$5x + 3y = 21 \quad \text{and} \quad 2x + 5y = 16$$

Answer: _____

6. An office places two orders from the same supplier. Four notebooks and three folders cost \$27. Six notebooks and five folders cost \$43. Let n be the price of one notebook in dollars and let f be the price of one folder in dollars. Write a system and solve it by elimination.

Answer: _____

7. Solve by elimination.

$$7x - 2y = 1 \quad \text{and} \quad 3x + 4y = 15$$

Answer: _____

8. A school buys pencils and pens by the box. Five boxes of pencils and two boxes of pens cost \$46. Three boxes of pencils and four boxes of pens cost \$50. Let p be the price of one box of pencils in dollars and let q be the price of one box of pens in dollars.

(a) Write a system and solve it by elimination. Give both prices in dollars.

Answer: _____

(b) Explain why multiplying the first equation by 2 does not change which pairs (p, q) solve the system.

9. Solve by elimination. Watch the signs — one of the answers is negative.

$$6x + 5y = 3 \quad \text{and} \quad 8x - 3y = 33$$

Answer: _____

10. A market sells cashews and almonds by the kilogram. Three kilograms of cashews and two kilograms of almonds cost \$57. Five kilograms of cashews and four kilograms of almonds cost \$101. Let c be the price of one kilogram of cashews in dollars and let d be the price of one kilogram of almonds in dollars.

(a) Write a system and solve it by elimination. Give both prices in dollars.

Answer: _____

(b) There are two ways to scale here: match the almond terms, or match the cashew terms. Describe both routes and justify which one takes fewer multiplications.